



ASANSOL ENGINEERING COLLEGE

Department - Computer Science & Engineering



PhishGuard

An Explainable Machine Learning Framework
for Real-Time Phishing URL Detection using
Feature Engineering and Ensemble Learning



Real-Time
Detection



Ensemble
Learning



Explainable AI
(SHAP)



Secure &
Reliable



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AGENDA

Roadmap of Our Presentation



PhishGuard

Detect. Explain. Protect.

- | | | | | | |
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PhishGuard aims to build a smarter and safer internet by combining accurate detection with explainable AI to fight against phishing threats.

Introduction

Phishing is a cyberattack where attackers create fake websites that imitate legitimate services to steal sensitive information such as passwords, banking credentials, OTPs and personal data.

Unlike malware, phishing relies primarily on social engineering and user trust.



HOW PHISHING ATTACK WORK?



PhishGuard is proposed as an intelligent phishing URL detection framework that uses **Machine Learning, Feature Engineering, and Ensemble Learning** to identify suspicious URL patterns. Instead of relying only on known malicious URLs, the system extracts meaningful characteristics from a URL, such as its length, special characters, domain structure, subdomains, IP-address usage, and other security-related properties.

The Growing Phishing Threat

Why Real-Time, Explainable URL Detection Matters

01 — Phishing remains a major attack vector

Phishing continues to be one of the most widely used techniques for credential theft, fraud and initial access.

02 — Attackers increasingly use deceptive URLs

Look-alike domains, URL manipulation, redirects and compromised websites make malicious links difficult to identify manually.

03 — Traditional detection has limitations

Rule-based and blacklist-based systems can struggle with newly created or previously unseen phishing URLs.

ATTACKER



Deceptive URL



VICTIM
clicks URL



Credential / Data
Theft

PHISHING IN 2025: KEY THREAT STATISTICS



36%
OF BREACHES
START WITH
PHISHING



\$4.88M
AVERAGE
BREACH COST



\$2.77B
BEC LOSSES
IN 2024



1,265%
SURGE IN
PHISHING EMAILS
SINCE GENERATIVE AI

⚠️ MANUAL INSPECTION

→ Unreliable for large-scale,
evolving threats



🛡️ PHISHGUARD

Automated • ML-Based

• Explainable Detection

THE GROWING IMPACT OF PHISHING ATTACKS

22.5%

Of all internet crimes
were phishing,
resulting in **\$70**
million in losses

13%

Year-over-year
increase kept
phishing as the most
common initial attack
vector in 2024

49%

Rise since 2021
linked to **AI-**
generated phishing
messages

PROBLEM STATEMENT

The Challenge of Detecting Phishing URLs



CORE CHALLENGE

How can we develop an **automated** and **explainable** system that learns suspicious URL patterns and detects phishing URLs **efficiently**, including previously unseen threats?



ADAPTIVE



AUTOMATED



EXPLAINABLE



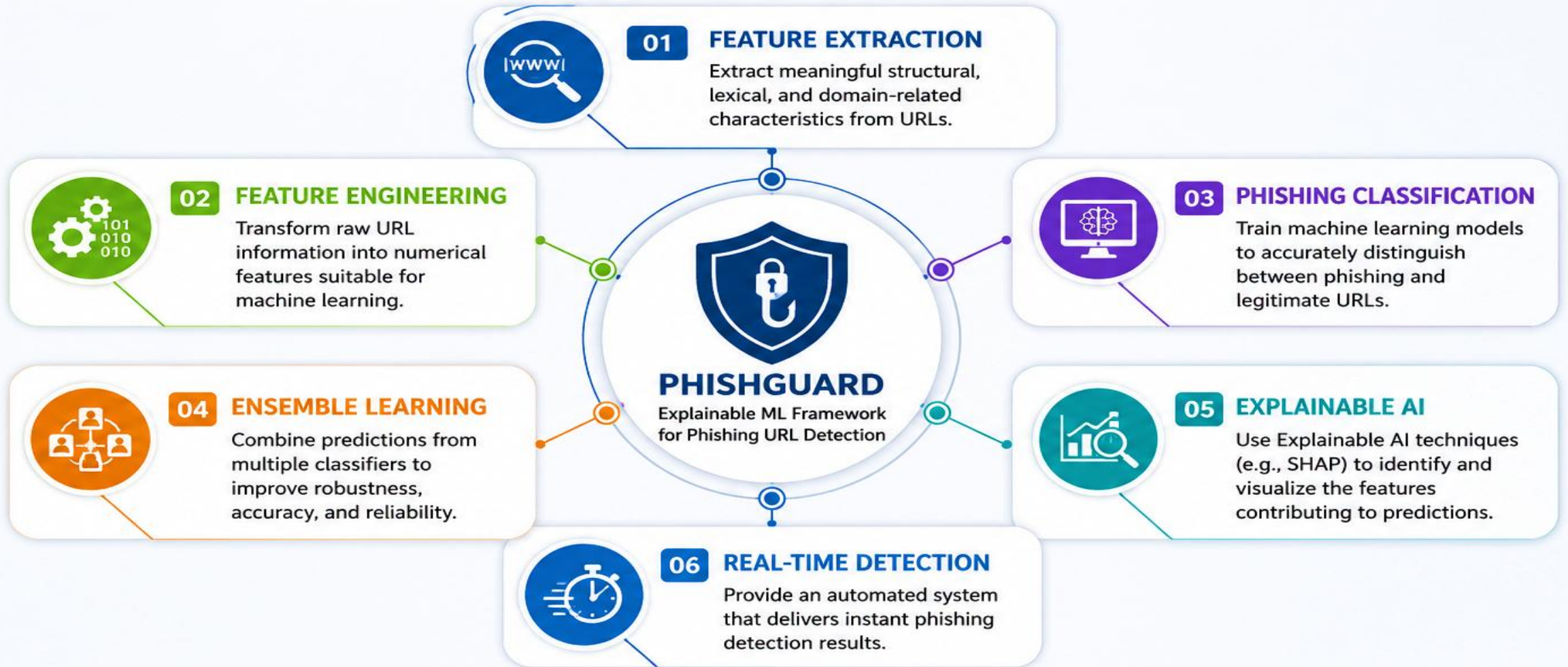
DEPENDABLE



NEED: An Adaptive + Machine Learning Based + Explainable Phishing Detection System

PROJECT OBJECTIVES

Objectives of PhishGuard



OVERALL GOAL

To build an Adaptive, Accurate, Automated, and Explainable System for Phishing URL Detection that enhances Online Safety.

EXPECTED OUTCOME



Detect
Phishing URLs



Provide
High Accuracy



Explain
Predictions



Maintain
History

EXISTING SYSTEM & ITS LIMITATIONS

Why Conventional Phishing Detection Approaches Are Not Sufficient

EXISTING APPROACHES

01 BLACKLIST-BASED DETECTION

- Checks whether a URL exists in a database of known malicious websites.
- Simple and easy to implement.

LIMITATION

Cannot reliably detect newly created or previously unseen phishing URLs.

02 RULE-BASED DETECTION

- Uses predefined rules such as suspicious characters, unusual domains, redirects or patterns.
- Fast and lightweight.

LIMITATION

Attackers can modify URL structures and bypass fixed rules easily.

03 REPUTATION-BASED DETECTION

- Uses domain reputation, trust scores and historical information.
- Helps identify suspicious domains.

LIMITATION

New domains may have insufficient reputation data.

04 MANUAL INSPECTION

- Security analysts manually examine URLs and website characteristics.
- Useful for detailed investigation.

LIMITATION

Time-consuming, dependent on experts, and not scalable.

KEY LIMITATIONS

LIMITED ZERO-DAY DETECTION

Previously unseen phishing URLs may not be present in existing databases or reports.

STATIC DETECTION

Fixed rules and signatures become ineffective as attackers continuously change their techniques.

SCALABILITY ISSUES

Manual analysis and rule updates cannot efficiently handle a large number of URLs.

LIMITED EXPLAINABILITY

Many systems only provide a simple "safe/malicious" result without explaining why.

LIMITATIONS
Conventional methods are insufficient against modern phishing threats.



CONVENTIONAL
APPROACHES



Need for intelligent,
adaptive and explainable
detection



PHISHGUARD

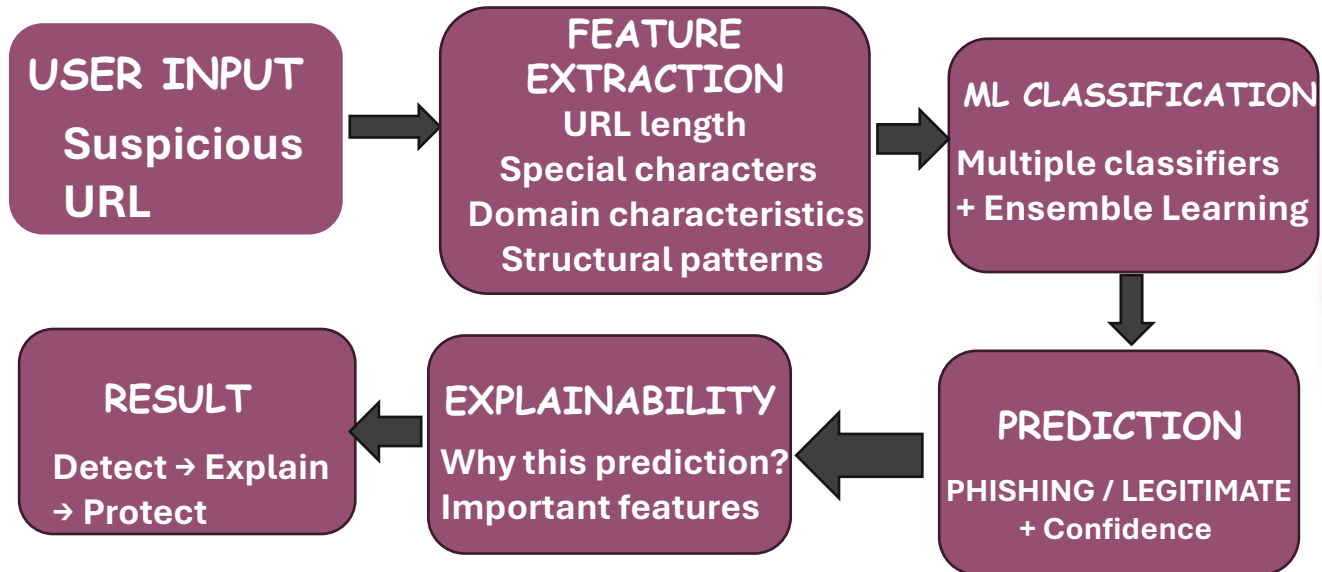
Machine Learning + Feature Engineering
+ Explainable Detection



Existing approaches are effective for known threats, but struggle with **new**, **evolving** and **obfuscated** phishing URLs.

Proposed Solution: PhishGuard

An Explainable Machine Learning Framework for Real-Time Phishing URL Detection



PhishGuard does not only detect a suspicious URL — it also explains the reasoning behind the prediction.

Core Approach:-

O1 – ANALYZE

Extract meaningful characteristics from the submitted URL.

O2 – PREDICT

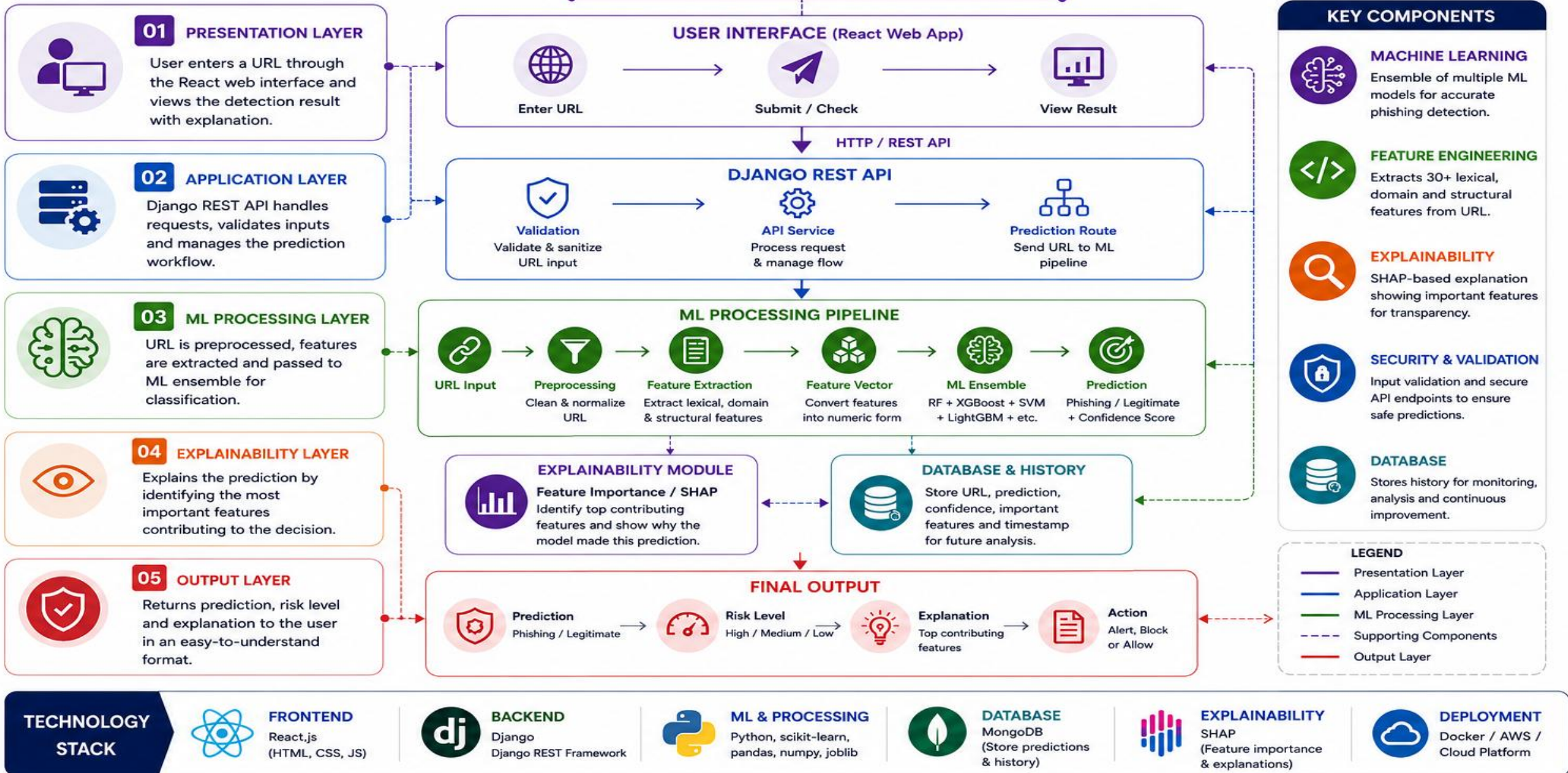
Use trained machine-learning models to classify the URL as phishing or legitimate.

O3 – EXPLAIN

Identify the URL characteristics that contributed to the prediction.

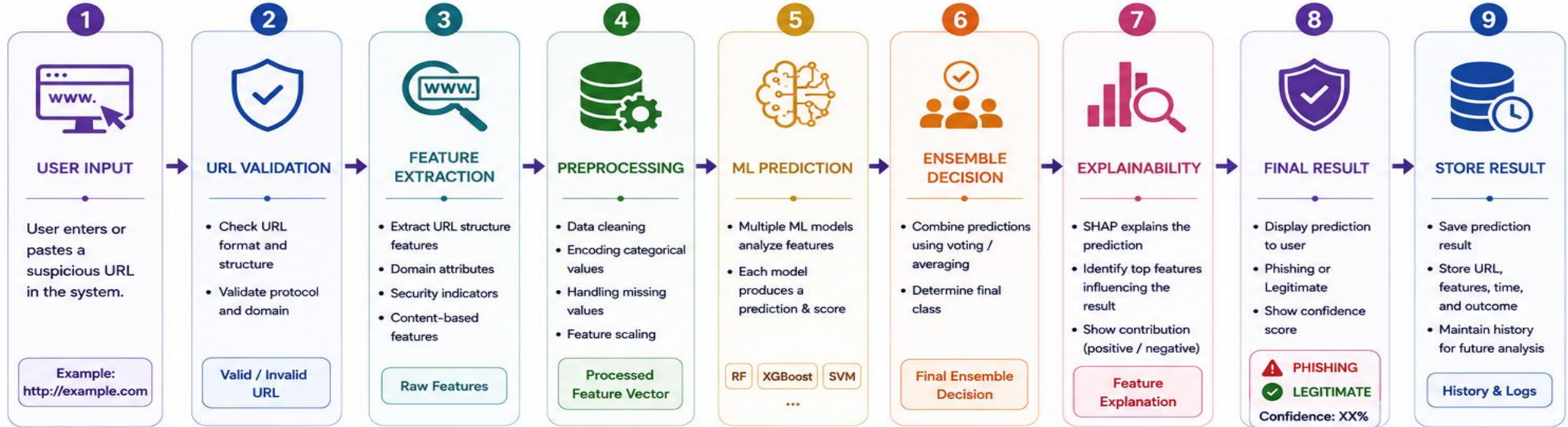
SYSTEM ARCHITECTURE

End-to-End Architecture of the PhishGuard Framework



PhishGuard Detection Workflow

End-to-End Process from URL Input to Explainable Result



WORKFLOW SUMMARY



Raw URL → Features → Models → Ensemble → Explanation → Result → Stored for Future Use

DATABASE (MongoDB)



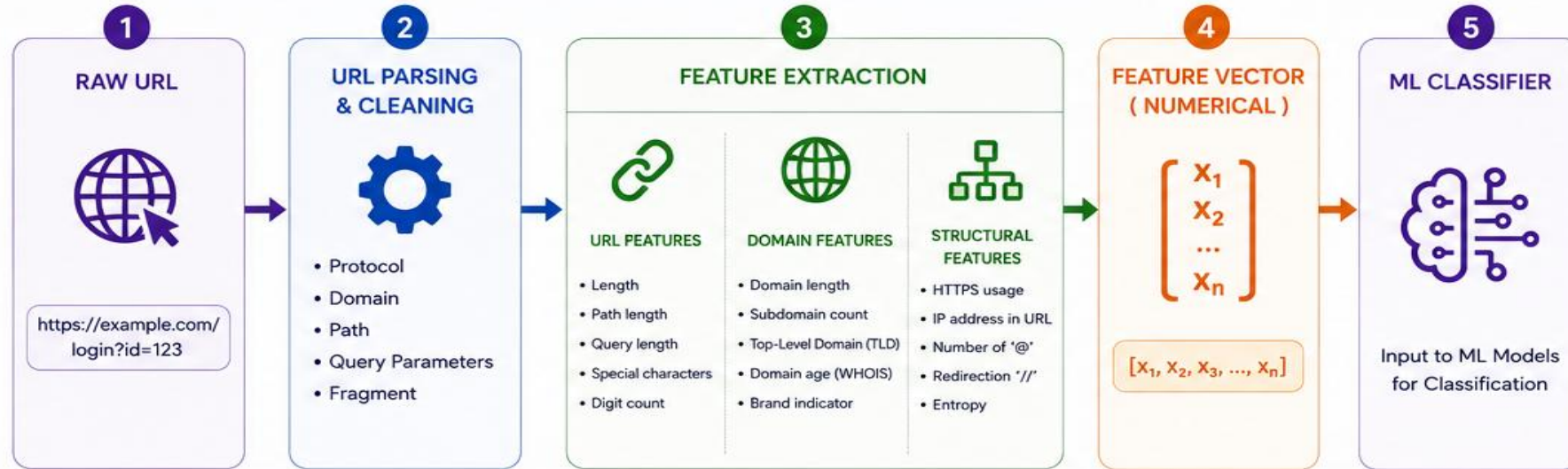
- Prediction History
- URL Records
- Feature Snapshots
- User Activity Logs
- Analytics & Insights



Key Takeaway: PhishGuard follows a systematic workflow that transforms a raw URL into an explainable and trustworthy security decision using feature engineering, machine learning, ensemble intelligence, and explainable AI.

URL FEATURE ENGINEERING

Converting Raw URLs into Machine-Learning Features



REPRESENTATIVE FEATURES	
CATEGORY	EXAMPLE FEATURES
URL STRUCTURE	• URL length • Path length • Query length
SPECIAL CHARACTERS	• Presence of @, -, _, ? • Count of //, =, ., %
DOMAIN FEATURES	• Domain length • Subdomain count • Top-Level Domain (TLD)
SECURITY FEATURES	• HTTPS usage • IP-based URL • Redirection count
COMPLEXITY FEATURES	• Digit count • Character count • Entropy score

PRACTICAL EXAMPLE

RAW URL

`https://google.com@malicious-site.com/login`



This URL appears like google.com but actually redirects to malicious-site.com after the '@' symbol.

EXTRACTED FEATURES (Sample)

has_@_symbol	url_length	subdomain_count	ip_based_url	https_used
@	High	2	IP	1
1			0	1

(Many more features are extracted to form the complete feature vector)

RESULT



PHISHING
High Risk

These numerical features are passed to the ML model which learns patterns and predicts whether the URL is **Phishing** or **Legitimate**.

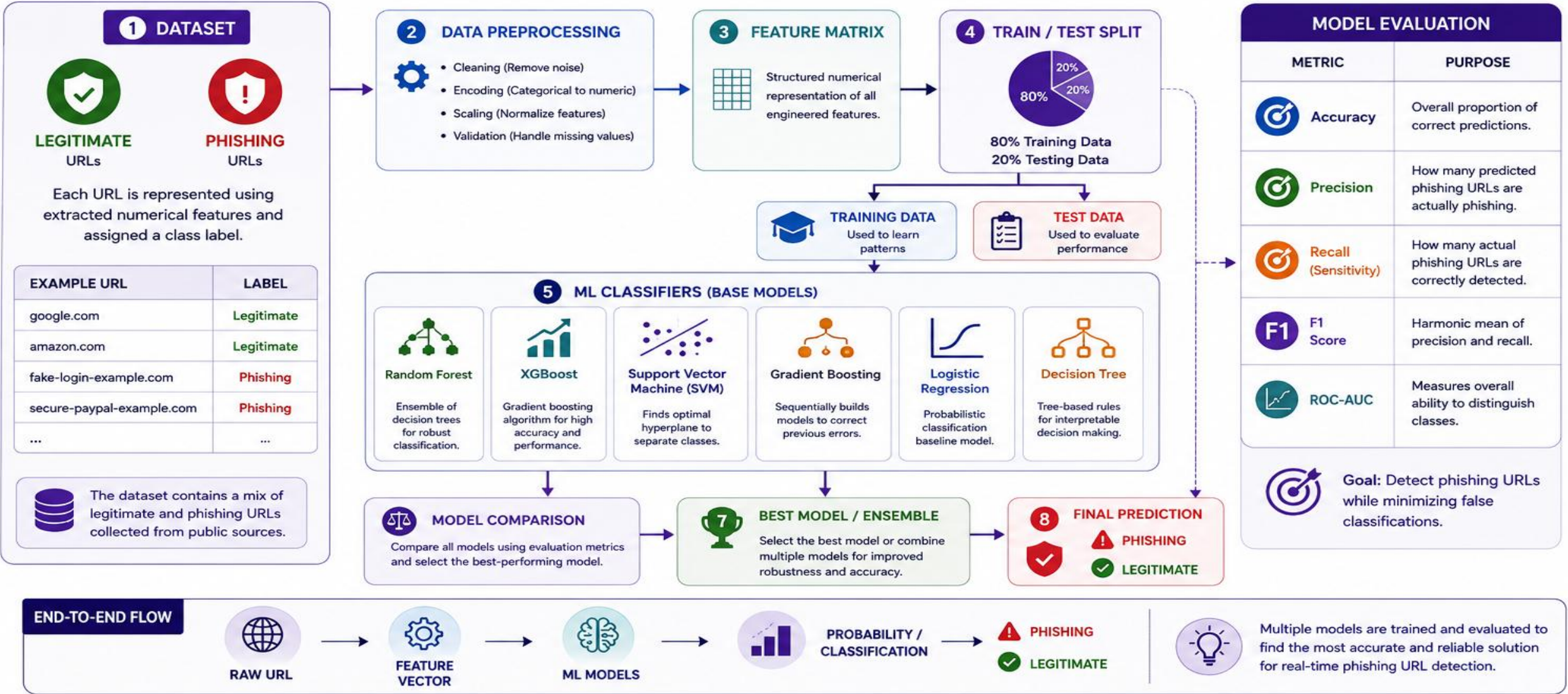


KEY TAKEAWAY

Feature engineering converts complex and unstructured URLs into meaningful numerical representations that help machine-learning models accurately distinguish between **phishing** and **legitimate** URLs.

MACHINE LEARNING MODEL DEVELOPMENT

Training and Evaluating Models for Phishing URL Classification



Key Takeaway:

We train multiple machine learning models on engineered features, evaluate them using standard metrics, and select the best model (or ensemble) to accurately classify URLs as **Phishing** or **Legitimate**.

EXPLAINABLE AI USING SHAP

Making Phishing Predictions Transparent and Interpretable

WHY EXPLAINABILITY?

Machine learning models can predict whether a URL is phishing or legitimate, but they often act like a “black box”.

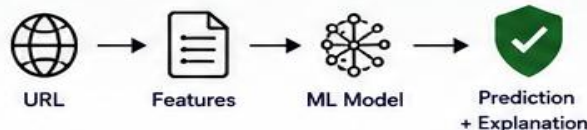
Explainability helps us understand **WHY** a URL was classified in a certain way.

Traditional ML (Black Box)



Prediction given, but the reason is unknown.

PhishGuard (Explainable AI)

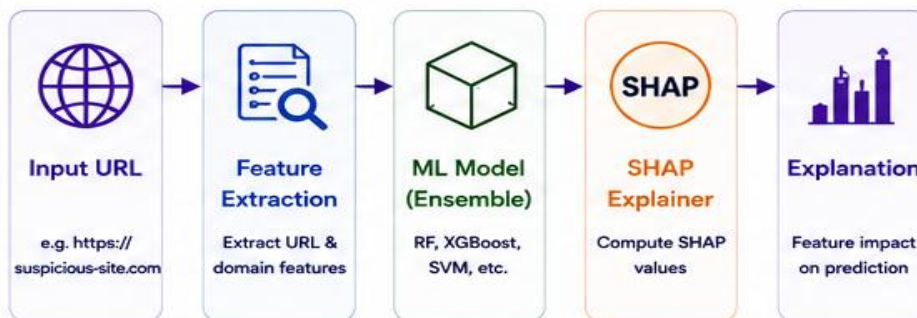


✓ Prediction along with understandable reasoning.



Explainability builds trust, improves user awareness, and helps in security analysis and model improvement.

SHAP EXPLAINABILITY PIPELINE



What is SHAP?

SHAP (SHapley Additive exPlanations) is a game-theoretic approach that assigns an importance value to each feature for a particular prediction.

Each feature pushes the prediction either towards **PHISHING** or towards **LEGITIMATE**.



Positive SHAP Value
(Increases phishing risk)



Negative SHAP Value
(Decreases phishing risk)

EXAMPLE SHAP EXPLANATION



PREDICTION: PHISHING

Confidence Score: 92%  92%

Top Contributing Features

Feature	Impact on Prediction	SHAP Value
URL Length	↑ 	+0.412
IP Address Used	↑ 	+0.358
@ Symbol Present	↑ 	+0.291
Subdomain Count	↑ 	+0.187
HTTPS Token	↓ 	-0.102
Presence of Dash	↑ 	+0.095

Decreases
Phishing Risk

← 0

→

Increases
Phishing Risk



SHAP values quantify how much each feature influenced the final prediction.



Key Takeaway:

SHAP makes PhishGuard's predictions transparent by highlighting the features that drive the decision. This helps users understand the risk and improves trust in automated phishing detection.

SYSTEM IMPLEMENTATION / USER INTERFACE

From URL Input to Explainable Results

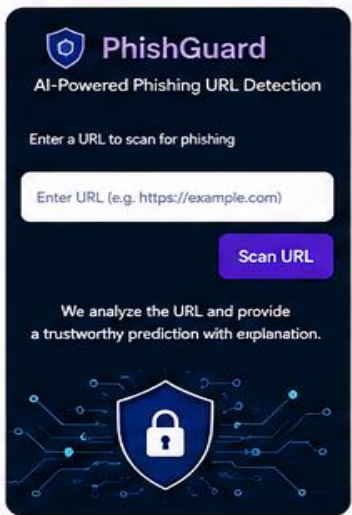
TECHNOLOGY STACK

-  **Frontend**
React.js (Vite)
-  **Backend**
Django + DRF
-  **ML Models**
scikit-learn, XGBoost, SVM, Random Forest
-  **Explainability**
SHAP
-  **Database**
MongoDB

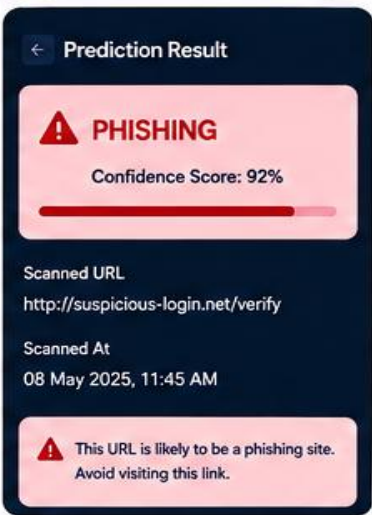
SYSTEM FLOW

- 1 User enters / pastes URL
- 2 Frontend sends request to Django API
- 3 Backend extracts features and preprocesses data
- 4 ML models generate predictions (Ensemble)
- 5 SHAP generates explanation of the prediction
- 6 Result is returned to UI and stored in database

1. URL INPUT PAGE



2. PREDICTION RESULT

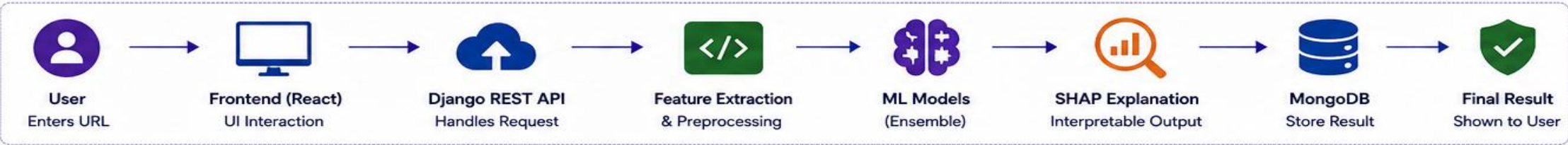


3. EXPLAINABILITY (SHAP)



4. HISTORY / DASHBOARD

URL	Result	Confidence	Time
suspicious-login.net	Phishing	92%	08 May 11:45
secure-pay-update.com	Phishing	87%	08 May 11:20
example.com	Legitimate	98%	08 May 10:55
mybank-secure.in	Phishing	90%	08 May 10:31
google.com	Legitimate	99%	08 May 09:50



KEY FEATURES



Real-time URL Scanning
Detect phishing URLs in real time.



High Accuracy
Ensemble learning for robust predictions.



Explainable Predictions
SHAP shows why a URL is classified as phishing.



Prediction History
Track and review all past predictions.



Secure & Reliable
Built with modern security best practices.



Responsive UI
Works seamlessly on all devices.



Key Takeaway: PhishGuard provides a seamless, secure and interpretable experience — from URL input to prediction and explanation.

RESULTS, EVALUATION & TESTING

Performance Analysis of PhishGuard

MODEL EVALUATION



Accuracy

95.24%

Overall correctness of the model



Precision

94.31%

How many predicted phishing URLs were actually phishing



Recall

96.17%

How many actual phishing URLs were correctly detected



F1-Score

95.23%

Harmonic mean of precision & recall



ROC-AUC

0.971

Model's ability to distinguish between classes

CONFUSION MATRIX

Actual / Predicted	Phishing	Legitimate
Phishing (Actual)	482 (TP)	18 (FN)
Legitimate (Actual)	22 (FP)	478 (TN)

Correct Predictions Incorrect Predictions

TP: True Positive | TN: True Negative
FP: False Positive | FN: False Negative

TESTING PERFORMED



Unit Testing

Tested individual modules and functions.



API Testing

Valid and invalid inputs, error handling, response validation.



Integration Testing

Frontend ↔ Backend ↔ ML Pipeline ↔ Database.



Performance Testing

Measured response time and system throughput.



Security Testing

Checked for input validation, SQL injection and other vulnerabilities.

DATASET INFORMATION



Dataset Source

Kaggle Phishing URL Dataset



Total Samples

10,000+ URLs



Classes

Phishing & Legitimate



Features Used

30+ Engineered Features



Dataset Split

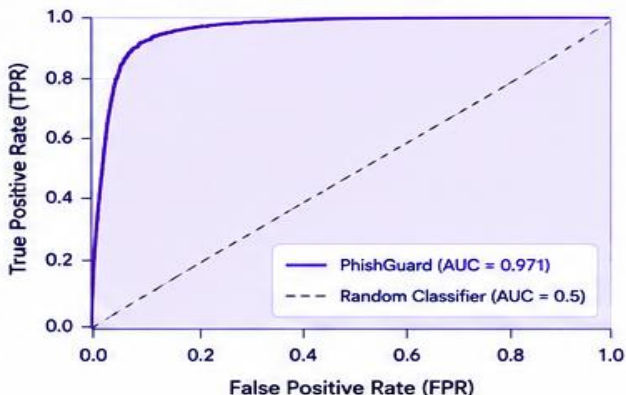
70% Training

15% Validation

15% Testing

These results are obtained using the **Ensemble model** (Random Forest + XGBoost + SVM) on the test dataset.

ROC CURVE (AUC)



CLASS DISTRIBUTION



SAMPLE PREDICTIONS

URL (Sample)	Actual Label	Predicted Label	Confidence	Result
http://secure-pay-update.com/login	Phishing	Phishing	92.4%	✓ Correct
https://www.google.com	Legitimate	Legitimate	98.7%	✓ Correct
http://free-gift-voucher.win	Phishing	Phishing	89.1%	✓ Correct
https://www.facebook.com	Legitimate	Legitimate	97.6%	✓ Correct
http://bit.ly/claim-prize-now	Phishing	Phishing	91.3%	✓ Correct



The model consistently classifies unseen URLs with high confidence.



Key Takeaway:

PhishGuard achieves high accuracy with a low false positive rate using an ensemble learning approach, robust feature engineering, and explainable AI to deliver reliable phishing detection.

CONCLUSION & FUTURE SCOPE

Towards a Safer Digital World with PhishGuard

CONCLUSION



PhishGuard

AI-Powered. Explainable. Reliable.

PhishGuard is an intelligent phishing URL detection system that combines feature engineering, ensemble machine learning, and explainable AI to identify malicious URLs with high accuracy and transparency.

By providing not only a prediction but also the reasons behind it, PhishGuard builds trust, enhances user awareness, and helps create a safer browsing experience.



Accurate Detection

High detection performance using an ensemble of robust ML models.



Explainable Results

SHAP explanations provide clear reasons for every prediction.



Real-time Protection

Fast and efficient detection to prevent users from phishing threats.



Data-driven Insights

Stores and analyzes prediction history for continuous improvement.



User-friendly Experience

Simple, responsive, and intuitive interface for all users.

FUTURE SCOPE

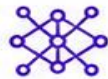
01



Advanced Feature Engineering

Incorporate webpage content analysis, DNS information, and behavioral features for better detection.

02



Deep Learning Models

Explore deep learning architectures (CNN, LSTM, Transformer) for improved accuracy.

03



Real-time Threat Intelligence

Integrate with live threat feeds and domain reputation services for up-to-date protection.

04



Browser Extension

Develop a browser extension for real-time phishing detection while users browse.

05



Interactive Explainability

Provide interactive SHAP visualizations for better user understanding.

06



Scalability & Deployment

Deploy as a scalable cloud service and support enterprise-level integration.

IMPACT



Protects Users

Helps users avoid phishing attacks and protects sensitive information.



Enhances Awareness

Explainable results educate users about phishing indicators.



Benefits Organizations

Reduces security risks, potential financial loss, and reputational damage.



Supports Cybersecurity

Contributes to building a safer and more secure digital ecosystem.



Together, let's build a smarter and safer internet for everyone.

“ Security is not a product, but a process.
Together, we can stay safe online. ”



PhishGuard
Detect. Explain. Protect.

THANK YOU